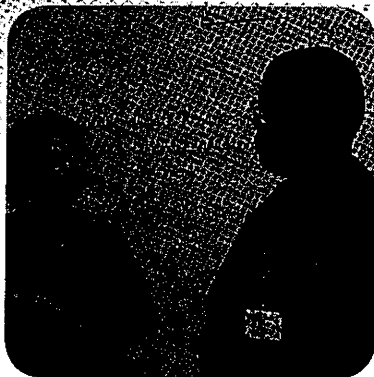


Self-Determination

2

Getting Students Involved in Leadership

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After completing this chapter,
the reader will be able to

- Explain what characterizes people who are self-determined
- Describe the role of educators in promoting self-determination
- Discuss some of the methods, materials, and instructional strategies that promote self-determination in the transition years
- Explain how efforts to promote self-determination contribute to student involvement with and progress in the general education curriculum
- Identify and utilize existing methods, materials, and strategies that promote student involvement in transition planning and decision making
- Explain why it is important to promote student involvement in transition planning and leadership development
- Describe how self-determination can be infused into 18–21 programs

Promoting and enhancing the self-determination of youth with disabilities has become recommended practice in transition services. The National Secondary Transition Technical Assistance Center identified self-determination as 1 of 16 evidence-based predictors of postschool employment, education, and independent living success (Test, Fowler, Richter, White, Mazzotti, Walker, & Korterling, 2009). This chapter defines *self-determination* and examines its importance to successful transition-related results. It then describes methods, materials, and strategies to promote self-determination and to support active student involvement in educational planning and decision making.

What Is Self-Determination?

The historical roots of self-determination for people with disabilities can be found in the normalization, independent living, disability rights, and self-advocacy movements and in legislative protections ensuring equal opportunities for people with disabilities (Ward, 1996). Promoting self-determination emerged as an instructional focus area in special education as a result of efforts to improve transition-related outcomes for youth with disabilities in the 1990s. As a result, there are numerous frameworks that can serve as a basis for instructional design, as well as specially designed instructional methods, materials, and strategies to promote self-determination (Wehmeyer, Abery, Mithaug, & Stancliffe, 2003; Wehmeyer et al., 2007; Wehmeyer & Field, 2007).

Although these frameworks propose different definitions of the self-determination construct, there is a general consensus in the field as to what characterizes *people* who are self-determined.

[Self-determined people] know how to choose—they know what they want and how to get it. From an awareness of personal needs, self-determined individuals choose goals, then doggedly pursue them. This involves asserting an individual's presence, making his or her needs known, evaluating progress toward meeting goals, adjusting performance, and creating unique approaches to solve problems. (Martin & Marshall, 1995, p. 147)

Our own work to promote self-determination, and the framework that guides our presentation of relevant interventions in this chapter, has been based on a functional model of self-determination in which self-determination is conceptualized as a dispositional characteristic (enduring tendencies used to characterize and describe differences between people; Wehmeyer, 2005; Wehmeyer et al., 2007) based on the *function* a behavior serves for an individual (Wehmeyer, 2005; Wehmeyer et al., 2007). *Self-determined behavior* refers to "volitional actions that enable one to act as the primary causal agent in one's life and to maintain or improve one's quality of life" (Wehmeyer, 2005, p. 117). Within this model, self-determined behavior refers to actions that are identified by four *essential characteristics*:

1. The person acted *autonomously*.
2. The behavior(s) are *self-regulated*.
3. The person initiated and responded to the event(s) in a *psychologically empowered* manner.
4. The person acted in a *self-realizing* manner.

These four essential characteristics describe the *function* of the behavior that makes it self-determined or not. That is, it is the function that the behavior serves for the individual that defines it as self-determined, not any specific class of behaviors themselves.

The concept of causal agency is central to our perspective. Broadly defined, *causal agency* implies that it is the individual who makes or causes things to happen in his or her life. One frequent misinterpretation of self-determination is that it simply means "doing it yourself." When self-determination is interpreted this way, however, there is an obvious problem for people with more severe disabilities, many of whom may have limits to the

Table 2.1. Component elements of self-determined behavior

Choice-making skills
Decision-making skills
Problem-solving skills
Goal-setting and -attainment skills
Independence, risk-taking, and safety skills
Self-observation, evaluation, and reinforcement skills
Self-instruction skills
Self-advocacy and leadership skills
Internal locus of control
Positive attributions of efficacy and outcome expectancy
Self-awareness
Self-knowledge

number and types of activities they can perform independently. However, the *capacity to perform* specific behaviors is secondary in importance to whether one is the causal agent (i.e., caused in some way to happen) over outcomes such behaviors are implemented to achieve. Thus, people who have severe physical disabilities can employ a personal assistant to perform routine activities and, if such functions are performed under the control of that person (i.e., the person with a disability), it is really a moot point whether the person physically performed the activity. Likewise, a person with a severe intellectual disability may not be able to "independently" (i.e., alone and with no support) make a complex decision, or solve a difficult problem. However, to the extent that supports are provided to enable that person to retain control over the decision-making process and to participate to the greatest extent in the decision-making or problem-solving process, he or she can become more self-determined.

Self-determination emerges across the life span as children and adolescents learn skills and develop attitudes that enable them to be causal agents in their lives. The *essential characteristics* that define self-determined behavior emerge through the development and acquisition of multiple, interrelated *component elements* (Table 2.1). Although not an exhaustive list, these component elements are particularly important to the emergence of self-determined behavior.

Self-Determination, Disability, and Empowerment

The role of educators in promoting self-determination is to teach students the knowledge and skills they need to become causal agents in their lives. However, it is important that educators not forget that the self-determination focus in disability services, including special education, emerged from deeply held convictions pertaining to the rights of people with disabilities to have a meaningful voice in their own lives. Within the context of the disability rights and advocacy movement, the self-determination construct has been imbued with an empowerment and "rights" orientation. *Empowerment* is a term usually associated with social and civil rights movements (Wehmeyer, 2004). *The American Heritage Dictionary of the English Language* (2004) states that the noun *empowerment* originally evolved to mean "to enable" or "to permit," and only recently has shifted to mean more control or power.

Although it is a contemporary buzzword, the word *empower* is not new, having arisen in the mid-17th century with the legalistic meaning "to invest with authority, authorize." Shortly thereafter it began to be used with an infinitive in a more general way meaning "to enable or permit." Its modern use originated in the civil rights movement, which sought political *empowerment* for its followers. Since people of all political persuasions have a need for a word that makes their constituents feel that they are or are about to become more in control of their destinies, *empower* has been adopted by conservatives as well as social reformers. (2000, pp. 586-587)

It is worth noting the meaning shift or drift that has occurred with use of the term since its 17th-century origination and the current linkages between empowerment and issues of control over one's life. It remains less than convincing, at least when the term is applied to the disability movement, that how many people use the term remains, in fact, closer to the original sense of the term: to authorize or invest with authority. The problem with that meaning with regard to people with disabilities is, of course, that in the end when one (and in our field, typically a professional) has the power to invest someone else with authority, one also has the power, presumably, to withhold granting that authority. Power and control remain, fundamentally, with the granter in that circumstance. In a similar way, the more current meaning identified by dictionary scholars (to enable or to permit) seems to offer two synonyms that, in the end, are not equally effective in solving the "granting authority" problem. That is, the act of "permitting" implies authority on the part of one person to allow another to do something, or not. The meaning of *empowerment* or, more accurately, *empower*, as meaning "to enable," is closer to the sense of the term as used when associated with social movements, particularly the disability rights movement, which typically uses the term in reference to actions that "enhance the possibilities for people to control their lives" (Rappaport, 1981, p. 15). *Enable* refers to providing opportunities and resources for something to happen.

Consideration of what it means to empower someone with a disability is more than just a semantic exercise. Well-intentioned professionals across many disciplines mistake empowerment as something that one grants or gives to someone else, to the end that the effort falls short of the standard of enhancing the possibilities for people to control their lives.

There is, as such, a bit of a catch-22 to issues pertaining to empowerment and professionals in transition and rehabilitation, in that many such professionals genuinely want to do whatever they can to empower people with disabilities, but similarly do not want to err in assuming that any ultimate authority to grant power or control lies within them. The way out of this conundrum is through efforts to *enable* people with disabilities to exert greater control in their lives—to become causal agents in their lives—and, as a function of such actions, to become empowered to do so to a greater extent. For professionals in transition education, the route to enablement is by providing opportunities and supports that promote and enhance the self-determination of students with disabilities.

Is Self-Determination Important to Transition Outcomes for Students with Disabilities?

Promoting the self-determination of students with disabilities has become a recommended practice in secondary education and transition services for several reasons. First, self-determination status has been linked to the attainment of more positive academic (Konrad, Fowler, Walker, Test, & Wood, 2007; Fowler, Konrad, Walker, Test, & Wood, 2007; Lee, Wehmeyer, Soukup, & Palmer, 2010) and transition outcomes, including more positive employment and independent living outcomes (Martorell, Gutierrez-Recacha, Pereda, & Ayuso-Mateos, 2008; Sowers & Powers, 1995; Wehmeyer & Palmer, 2003; Wehmeyer & Schwartz, 1997), and more positive quality of life and life satisfaction (Lachapelle et al., 2005; Nota, Ferrari, Soresi, & Wehmeyer, 2007; Shogren, Lopez, Wehmeyer, Little, & Pressgrove, 2006; Wehmeyer & Schwartz, 1998).

Second, research across special education disability categories has established the need for intervention to promote self-determination, documenting that students with intellectual disability (Wehmeyer et al., 2007), learning disabilities (Field, Sarver, & Shaw, 2003; Pierson, Carter, Lane, & Glaeser, 2008), emotional and behavioral disorders (Carter, Lane, Pierson, & Glaeser, 2006; Pierson et al., 2008), and autism (Wehmeyer, Shogren,

Zager, Smith, & Simpson, 2010) are less self-determined than their nondisabled peers. Teachers believe that teaching students to become more self-determined is important (Carter, Lane, Pierson, & Stang, 2008; Thoma, Pannozzo, Fritton, & Bartholomew, 2008; Wehmeyer, Agran, & Hughes, 2000) and there are numerous curricular and instructional models identified to enable them to provide this instructional focus (Test, Karvonen, Wood, Browder, & Algozzine, 2000; Wehmeyer & Field, 2007). In a meta-analysis of single-subject and group-subject design studies, Algozzine, Browder, Karvonen, Test, and Wood (2001) found evidence for the efficacy of instruction to promote component elements of self-determined behavior, including self-advocacy, goal setting and attainment, self-awareness, problem-solving skills, and decision-making skills. Cobb, Lehmann, Newman-Gonchar, and Alwell (2009) conducted a narrative metasynthesis—a narrative synthesis of multiple meta-analytic studies—covering seven existing meta-analyses examining self-determination and concluded that there is sufficient evidence to support the promotion of self-determination as effective. Also, research documents the positive impact of efforts to promote student involvement in educational and transition planning (Martin, Van Dycke, Christensen, Greene, Gardner, & Lovett, 2006; Mason, Field, & Sawilowsky, 2004; Test et al., 2004) on more positive transition and self-determination-related outcomes.

Third, and importantly, causal evidence of the impact of interventions to promote self-determination on student self-determination status is increasingly available. Wehmeyer, Palmer, Shogren, Williams-Diehm, and Soukup (in press) conducted a randomized trial control group study of the effect of interventions to promote self-determination on the self-determination status of high school students receiving special education services under the categorical areas of intellectual disability and learning disabilities. Students in the treatment group ($n = 235$) received instruction using a variety of instructional methods to promote self-determination and student involvement in educational planning meetings over 3 years, while students in the control group ($n = 132$) received no such intervention. The self-determination of each student was measured with two instruments, *The Arc's Self-Determination Scale* (SDS; Wehmeyer & Kelchner, 1995) and the *AIR Self-Determination Scale* (AIR; Wolman, Campeau, Dubois, Mithaug, & Stolarski, 1994; both described subsequently) across three measurement intervals. Wehmeyer and colleagues found that students with intellectual disability and learning disabilities who participated in interventions to promote self-determination over a 3-year period showed significantly more positive patterns of growth in their self-determination scores than did students not exposed to interventions to promote self-determination during the same time period.

In summary, there is an expanding base of evidence suggesting that higher self-determination and increased capacity in the component elements of self-determined behavior results in better transition-related outcomes for youth and young adults with disabilities. The obvious next issue is how to achieve this important outcome.

Assessments, Methods, and Strategies to Promote Self-Determination

Efforts to enhance the self-determination of youth with disabilities should involve multiple, parallel activities focused on teaching skills related to the component elements of self-determined behavior and promoting active involvement in educational planning and decision making. We have already highlighted research establishing that students can acquire skills and knowledge pertaining to self-determination if provided instruction and that such instruction can positively impact transition-related outcomes for youth and young adults with disabilities. In the context of school reform efforts, such efforts take on even greater urgency.

Importance of Self-Determination to Access to the General Education Curriculum

The 1997 amendments to the Individuals with Disabilities Education Act (IDEA 1997) and their associated regulations included statutory and regulatory language intended to ensure that students with disabilities had access to the general curriculum. Section 300.347(a)(3) in IDEA 1997 required that the individualized education program (IEP) of students with disabilities include

[a] statement of the special education and related services and supplementary aids and services to be provided to the child, or on behalf of the child, and a statement of the program modifications or supports for school personnel that will be provided for the child

- (i) to advance appropriately toward attaining the annual goals;
- (ii) to be involved and progress in the general curriculum;
- (iii) to be educated and participate with disabled and non-disabled children.

In fact, as reflected in the language in part (ii), what IDEA required was that students with disabilities be involved with and show progress in the general curriculum. The term "access to the general curriculum" refers to this requirement for student involvement and progress. The general curriculum was defined in the regulations as referring to "the same curriculum as for nondisabled children" (*Federal Register*, 1999, p. 12,592). The intent of these access provisions was threefold, as described by U.S. Department of Education officials:

1. That all students, including students with disabilities, would have access to a challenging curriculum
2. That all students, including students with disabilities, would be held to high expectations
3. To align special education practice with accountability mechanisms emerging through school reform efforts

The 2004 reauthorization of IDEA (IDEA 2004) contained all of the original IDEA 1997 mandates and added several new requirements, including that schools ensure that the IEP team includes someone knowledgeable about the general education curriculum and that the team meet at least annually to address any lack of expected progress in the general education curriculum. IDEA 2004 also changed the term to general *education* curriculum. Finally, the regulations to IDEA 2004 prohibited a student with a disability from being removed from the general education setting based solely upon needed modifications to the general education curriculum.

Spooner, Dymond, Smith, and Kennedy (2006) reviewed the approaches to promoting the access of students with disabilities, identifying a focus on self-determination as one of only a few such approaches. Wehmeyer, Field, Doren, Jones, and Mason (2004) identified two ways in which promoting self-determination will, in fact, promote access to the general education curriculum. First, state and local standards frequently include goals and objectives that pertain to component elements of self-determined behavior. By identifying where in the general curriculum all students are expected to learn skills and knowledge related to the component elements of self-determined behavior, teachers can promote self-determination and promote progress in the general education curriculum. Second, model processes that identify how to promote access to the general education curriculum emphasize the importance of curriculum modifications, adaptations, and augmentations that enable students to interact with curricular content. Teaching students the skills that enable them to be more self-determined, such as goal setting and attainment, problem solving, self-regulation and self-management, self-directed learning, coping and organizational, and leadership and teamwork skills will also enable them to more effectively interact with the general education curriculum. These are discussed subsequently.

There is now a clear evidence base that teaching students to self-regulate learning or teaching students self-directed learning strategies such as self-monitoring or self-instruction has beneficial outcomes for students with severe disabilities in educational goal attainment, including goals linked to transition-related outcomes and to the general education curriculum (Agran, Cavin, Wehmeyer, & Palmer, 2006; Agran, Wehmeyer, Cavin, & Palmer, 2008, 2010). Recently, Wehmeyer and colleagues have provided evidence that teaching students to self-regulate learning using the *Self-Determined Learning Model of Instruction* (SDLMI; Wehmeyer et al., 2000) results in the attainment of goals linked to the general education curriculum (Lee et al., 2010). Shogren, Palmer, Wehmeyer, Williams-Diehm, and Little (in press) conducted a group-randomized trial control group study examining the impact of intervention using the SDLMI on student academic and transition goal attainment and on access to the general education curriculum for students with intellectual disability and learning disabilities, showing the efficacy of the model for both goal attainment and access to the general education curriculum.

Assessment in Self-Determination

As is the case with instruction in any content area, assessment and instruction go hand in hand in efforts to promote self-determination. Determining instructional and curricular needs in the area of self-determination will involve a combination of standardized and informal procedures incorporating input from multiple sources, including the student, his or her family, professionals, and others. Informal procedures will be similar to those described by Clark (1996) with regard to transition assessment. Clark identified informal assessment from which transition-related decisions can be made as including

- a. Situational or observational learning styles assessments
- b. Curriculum-based assessment
- c. Observational reports from teachers, employers, and family members
- d. Situational assessments in home, community, and work settings
- e. Environmental assessments
- f. Personal future-planning activities
- g. Structured interviews with students
- h. Structured interviews with parents, guardians, advocates, or peers
- i. Adaptive, behavioral, or functional skill inventories
- j. Social histories
- k. Employability, independent living, and personal social skills rating scales
- l. Technology or vocational education skills assessments.

These types of assessment procedures enable planners to form a complete picture of student needs, interests, and abilities by gathering input from multiple sources, and are important for determining the same things as they pertain to the need for instruction to promote self-determination.

Norm-Referenced Measures of Self-Determination

The Arc's Self-Determination Scale The SDS (Wehmeyer & Kelchner, 1995; available online at <http://www.ou.edu/content/education/centers-and-partnerships/zarrow/self-determination-assessment-tools.html>) is a 72-item self-report measure based on the functional theory of self-determination. A total of 148 points are available on the scale, with higher scores indicating higher levels of self-determination. An overall self-determination score, as well as subscale scores for each of the four essential characteristics of self-determined behavior—autonomy, self-regulation, psychological empowerment,

and self-realization—can be calculated. The SDS was developed and normed with 500 adolescents with intellectual disability and learning disabilities and was demonstrated to have adequate reliability and validity (Wehmeyer, 1996). Subsequent research (Shogren et al., 2008) has verified the proposed theoretical structure of the SDS (i.e., four related, but distinct, subscales that contribute to a higher-order self-determination construct).

The SDS has been used to conduct research into the relationship between self-determination and positive adult outcomes (Wehmeyer & Schwartz, 1997) and quality of life variables (Wehmeyer & Schwartz, 1998) and the relationship between self-determination and environmental factors (Wehmeyer & Bolding, 1999, 2001) and to validate instructional strategies to promote self-determination (S. Lee et al., 2010; Y. Lee et al., 2011; Shogren et al., 2010; Wehmeyer, Palmer, Agran, Mithaug, & Martin, 2000; Wehmeyer, Shogren, et al., in press) and materials to promote student-directed transition planning (Wehmeyer & Lawrence, 1995; Wehmeyer, Palmer, Lee, Williams-Diehm, & Shogren, 2011).

One potential use of the SDS is to generate discussion about items the student finds interesting or problematic or wants to discuss more broadly. A second use of the SDS involves scoring it and comparing total, domain, and subdomain scores with scale norms and, more importantly, examining individual strengths and weaknesses across the domains. Less than optimal performances in any area of the SDS should be followed by learning opportunities and experiences that enable the student to make progress in that particular area, and any use of the SDS with individual students should focus on potential educational goals and objectives. This discussion, in turn, can consider possible educational programs and activities to address and meet these goals and objectives.

The AIR Self-Determination Scale The AIR Self-Determination Scale (AIR; Wolman et al., 1994; available online at <http://www.ou.edu/content/education/centers-and-partnerships/zarrow/self-determination-assessment-tools.html>) assesses student capacity and opportunity for self-determination. The AIR has a Student (AIR-S), Educator, and Parent version. The AIR-S has 24 questions and also yields capacity and opportunity subscale scores. The capacity subscale consists of questions related to things students do related to self-determination and how students feel about performing these self-determined behaviors. The opportunity subscale consists of questions regarding students' perceptions of their opportunities to perform self-determined behaviors at home and at school.

The AIR was developed and normed with 450 students with and without disabilities in California and New York (Wolman et al., 1994). The AIR was demonstrated to have adequate reliability and validity in the measurement of capacity and opportunity for self-determination (see Wolman et al., 1994, for details). Recent research (Shogren et al., 2008) has confirmed the theoretical structure of the AIR Self-Determination Scale (i.e., two related subscales—capacity and opportunity—that contribute to a higher-order self-determination construct). This research also confirmed that, although the SDS and the AIR-S are related ($r = .50$), they are measuring distinct aspects of the self-determination construct.

Self-Determination Assessment Battery¹ The Self-Determination Assessment Battery was developed by Field, Hoffman, and Sawilowsky (2004). It contains five instruments that measure cognitive, affective, and behavioral factors related to self-determination. In addition, these factors are assessed from the perspectives of the student, the teacher, and the parent. This battery of instruments was developed to assess knowledge, behavior, and affective components of self-determination from these varied perspectives and within the context of an intervention theory to promote self-determination forwarded by Field and Hoffman (1994). The five instruments in the battery are as follows (Field et al., 2004):

¹Information on ordering the Self-Determination Assessment Battery can be obtained from the Wayne State University Center for Self-Determination and Transition Business Office at (313) 577-1638 or sdtalk@wayne.edu.

1. *Self-Determination Knowledge Scale (SDKS) Pretest*: The SDKS-pre- and SDKS-posttest are each 37-item structured response instruments for use with students with mild-moderate intellectual disability and learning disabilities, designed to assess their cognitive knowledge of self-determination skills as taught in the Field and Hoffman (1994) *Steps to Self-Determination* curriculum.
2. *Self-Determination Parent Perception Scale (PPS)* and
3. *Teacher Perception Scale (TPS)*: The PPS and TPS are 30-item questionnaires administered to parents and teachers, respectively. The items in these questionnaires were also derived from the Field and Hoffman (1994) model of self-determination intervention. The teacher or parent rates the student on a variety of behaviors, abilities, and skills associated with self-determination.
4. *Self-Determination Observation Checklist (SDOC)*: The SDOC is a 38-item behavioral observation checklist designed to be administered by classroom teachers or other appropriate personnel in the school environment. The student is observed for approximately 5 minutes during a class period. Behaviors that correlate to self-determination are checked.
5. *Self-Determination Student Scale (SDSS)*: The SDSS is a 92-item self-report instrument that measures both affective and cognitive aspects of the student's self-determination. The items contain a brief stimulus, to which the student marks "That's me" or "That's not me." The SDSS yields a variety of subscale scores; the general subscales relate to a student's sense of global self-determination, whereas the specific subscales relate primarily to application in their education, home, and related environmental settings. The positive subscales indicate self-determination in areas of perceived strength, and the negative subscales indicate areas of perceived weakness in self-determination.

The Self-Determination Assessment Battery instruments have many possible uses in education. First, they can be used to assist in educational planning by identifying areas of similarity and discrepancy among the three perspectives of the student, teacher, and parent. This provides an opportunity for discussion among the student, teacher, and parent to determine the reasons for this discrepancy. For example, it may be that a student is exhibiting skills in the home that he or she is not displaying at school or it may be that the teacher and the parent were using different criteria to evaluate the student's performance. Because students are not only evaluated from different perspectives but also being assessed in three different areas (cognition/knowledge, behavior, and affect), examining the differences in the three different areas can also help to determine appropriate interventions.

All of these instruments clearly have varied uses for educational planning, both as discussion tools in educational planning meetings that can help to promote greater self-awareness and as tools that can help to identify appropriate educational interventions. In addition, the instruments can be used for program evaluation or research purposes. By using the instruments as pre- and posttests before and after an instructional intervention, data can be obtained that can help to assess the effectiveness of the intervention.

Infusing Instruction to Promote Self-Determination in the General Education Curriculum

Infusing instruction on component elements of self-determined behavior (Table 2.1) into instruction across content areas provides the first focus for promoting self-determination. This section briefly identifies some key strategies to achieve this instructional outcome.

Goal Setting

Goal-setting and attainment skills are critical to students with disabilities becoming more self-determined. Goals specify what a person wishes to achieve and act as regulators of

human behavior. If a person sets a goal, then it increases the probability that he or she will perform behaviors related to that goal (Latham & Locke, 1991). The process of promoting goal-setting and attainment skills involves teaching students to

1. Identify and define a goal clearly and concretely
2. Develop a series of objectives or tasks to achieve the goal
3. Specify the actions necessary to achieve the desired outcome

Goal-setting activities can be easily incorporated into a variety of transition-related activities and across multiple instructional areas, as well as in the educational planning process, including through student-directed planning activities such as those discussed subsequently.

Research has suggested some general strategies to follow to make goals both meaningful and attainable for students with disabilities. First, goals should be challenging for the student. They should not be so challenging that the student cannot attain them, as this will lead to frustration and withdrawal from participation, but they must provide enough motivation for the student to work to attain them. If goals are too easy, then there is neither motivation to engage in the work necessary to attain them nor a feeling of accomplishment after achieving them. Although it is preferable for students to participate in setting their own goals at whatever level is appropriate given the nature of their disability, if this is not possible and goals need to be set by teachers, then the student's preferences and interests should be incorporated into the goal to increase the student's motivation to pursue the goal. Goals that have personal meaning are more likely to be attained (Doll & Sands, 1998).

Choice Making

Choice making (e.g., the expression of a preference between two or more options) has been linked to multiple outcomes of benefit to transition. There is an emerging database showing that incorporating choice-making opportunities into interventions to reduce problem behaviors of children and youth with disabilities results in improved behavioral outcomes (Shogren, Faggella-Luby, Bae, & Wehmeyer, 2004). Cooper and Browder (1998) found that teaching young adults to make choices improved outcomes of **community-based instruction**. Watanabe and Sturmey (2003) found that promoting choice-making opportunities in vocational tasks for young adults with disabilities increased engagement in the activity.

In addition, by making choices, students, particularly younger children, learn that they can exert control over their environment. For students to fully understand the process of choice making, including the various effects of making certain choices, choices need to be real and meaningful. A variety of adaptive equipment, ranging from picture communication systems to computer technology, can be used to support students with more severe disabilities to indicate their preferences. Such strategies may be particularly helpful for students with autism, given their preference for information in a concrete, visual form. Choice opportunities can and should be infused through the school day. Students can be provided opportunities to choose within or between instructional activities. They can also choose with whom they engage in a task, where they engage in an activity, and whether they complete an activity (Wehmeyer et al., 2007).

Problem Solving

A problem is an activity or task for which a solution is not known or readily apparent. The process of solving a problem involves (D'Zurilla & Goldfried, 1971, pp. 107-126):

1. Identifying and defining the problem
2. Listing possible solutions

3. Identifying the impact of each solution
4. Making a judgment about a preferred solution
5. Evaluating the efficacy of the judgment

Developing effective social problem-solving skills is central to the process of becoming self-determined. These skills are central to a student's capacity to interact with other people and to cope with problems that arise in social contexts.

Limitations in social problem-solving skills have been linked to difficulties in employment, community, and independent living situations for people with **developmental disabilities** (Gumpel, Tappe, & Araki, 2000). Storey (2002) reviewed the empirical literature pertaining to improving social interactions for workers with disabilities and determined that problem-solving skills contributed to more positive workplace social interactions. O'Reilly, Lancioni, and O'Kane (2000) found that incorporating instruction in problem solving into social skills instruction improved employment outcomes for supported workers with traumatic brain injuries.

A number of strategies to promote problem solving have been evaluated for students with disabilities. Bauminger (2002, 2007a, 2007b) developed a curriculum to teach students with autism social and interpersonal problem-solving skills. Students were taught about social concepts, such as starting a conversation, and then were presented a vignette of a student having difficulty implementing the skill. Students went through an eight-stage problem-solving process with their teacher in which they

1. Defined the problem
2. Discussed the emotions associated with the problem
3. Defined the alternative social actions
4. Considered the consequences of each alternative
5. Made a decision about the best alternative
6. Role-played the solution with their teacher
7. Received homework to practice the social skill covered in the lesson at home with peers
8. Received feedback from the teacher on the homework

After 7 months, students generated more appropriate solutions to problems faced in social situations and initiated more social interactions with their peers.

Bernard-Opitz, Sriram, and Nakhoda-Sapuan (2001) developed a computer program to teach students with developmental disabilities social problem-solving skills. The program first presented pictures or videos of people experiencing social conflicts. The program guided students through an animated problem-solving process in which they were asked to generate alternative solutions. After identifying an alternate solution, a video clip of the actors resolving the problem was presented. As students had repeated experience with the program, they generated more alternative solutions. Bernard-Ripoli (2007) used video self-modeling as a strategy, combined with social stories, to assist a child with Asperger syndrome understand emotions pertaining to social interactions and social problem solving. In addition, problem-solving instruction is a component of many self-regulation strategies, discussed subsequently.

Decision Making

A decision-making process involves coming to a judgment about which solution is best at a given time. Making effective decisions typically involves (Beyth-Marom, Fischhoff, Quadrel, & Furby, 1991, pp. 19–59)

1. Identifying alternative courses of action
2. Identifying the possible consequences of each action

3. Assessing the probability of each consequence occurring
4. Choosing the best alternative
5. Implementing the decision

Although the ability to engage in this process develops with age, research has shown that young children can engage in a systematic decision-making process, often by reducing and simplifying the steps in the decision-making process, although they are not as effective as older students (Crone, Vendel, & van der Molen, 2003). Thus, working to promote systematic decision-making skills is best addressed at the secondary level, whereas at the elementary level, a focus on choice making and problem solving can support the development of effective decision-making skills later in life.

Studies have shown repeatedly that adolescents with disabilities can effectively participate in the decision-making process (Wehmeyer et al., 2007), want to be involved in decisions related to their life (Ruef & Turnbull, 2002), and benefit from instruction to do so. Teaching young women with intellectual disabilities to make more effective decisions improved their capacity to identify potentially abusive social interactions (Khemka, 2000). Datillo and Hoge (1999) found that teaching decision making to adolescents with intellectual disability in the context of a leisure education program improved their acquisition of socially valid leisure knowledge and skills.

To support students with disabilities to acquire decision-making skills, a number of strategies can be implemented throughout the student's educational career. Early on, students should be provided a wide array of choice opportunities and receive instruction regarding how to make effective choices, as discussed previously. As students age, they should be provided overt instruction in the decision-making process. When teaching decision-making skills, opportunities to make decisions should be embedded in the curriculum. By supporting students to make decisions in "real-world" situations, they will better develop their ability to conceptualize and generalize the decision-making process. The Virginia Department of Education's Self-Determination Project (<http://www.indetermined.org>) provides free lesson plans and templates for student involvement in the IEP.

Self-Regulation and Student-Directed Learning

Each of the aforementioned areas is important to enable students to self-regulate their behavior and their lives. Self-regulation is the process of setting goals, developing action plans to achieve those goals, implementing and following the action plans, evaluating the outcomes of the action plan, and changing action plans if the goal was not achieved (Mithaug, Mithaug, Agran, Martin, & Wehmeyer, 2003). The skills associated with self-regulation enable students to examine their environments, evaluate their repertoire of possible responses, and implement and evaluate a response (Whitman, 1990).

Student-directed learning strategies involve teaching students strategies that enable them to modify and regulate their own behavior (Agran, King-Sears, Wehmeyer, & Copeland, 2003). The emphasis in such strategies is shifted from teacher-mediated and -directed instruction to student-directed instruction. Research in education and rehabilitation has shown that student-directed learning strategies are as successful—and often more successful—as teacher-directed learning strategies, and these strategies are effective means to increase independence and productivity. A variety of strategies has been used to teach students with disabilities how to manage their own behavior or direct learning. Among the most commonly used strategies are picture cues and antecedent cue regulation strategies, self-instruction, self-monitoring, self-evaluation, and self-reinforcement. These are briefly introduced next (see Agran et al., 2003, for a comprehensive treatment of student-directed learning strategies).

Picture cues and antecedent cue regulation strategies involve the use of visual or audio cues that students use to guide their behavior. Visual cues typically involve photographs, illustrations, or line drawings of steps in a task that support students to complete an activity

that consists of a sequence of tasks. Audio cues include prerecorded, taped directions or instructions that the students can listen to as they perform a task. Emerging technologies, such as handheld computers, provide new and potentially powerful vehicles to deliver visual or auditory cues to learners. Picture cues and antecedent cue regulation strategies have been used to teach individuals with intellectual disability complex work task sequences and to promote on-task behavior and independent work performance (Agran et al., 2003; Mithaug et al., 2003).

Self-instruction involves teaching students to provide their own verbal cues before the execution of target behaviors. Students and adults with intellectual disability have been taught to use self-instruction to solve a variety of work problems, to complete multistep sequences, and to generalize responding across changing work environments (Wehmeyer et al., 2007). Graham and Harris (1989) found that a self-instructional strategy improved the essay composition skills of students with learning disabilities.

Self-monitoring involves teaching students to observe whether they have performed a targeted behavior and whether the response met whatever existing criteria were present. Teaching students self-monitoring strategies has been shown to improve critical learning skills and classroom involvement skills of students with severe disabilities (Agran et al., 2005; Hughes et al., 2002). Woods and Martin (2004) found that teaching supported employees to self-manage and self-regulate work tasks improved employers' perceptions of the employee and improved work performance.

Self-evaluation and *self-reinforcement* involve teaching the students to compare their performance (as tracked through self-monitoring) with a desired goal or outcome and to administer consequences to themselves (e.g., verbally telling themselves they did a good job). Self-reinforcement allows students to provide themselves with reinforcers that are accessible and immediate. Given access to self-administered reinforcement, behavior change may be greatly facilitated and both procedures have been shown to improve generalization of learning (Agran et al., 2003).

Self-Advocacy

Students with disabilities need to learn the skills to advocate on their own behalf. To be an effective self-advocate, students have to learn both how to advocate and what to advocate. There are ample opportunities for students to practice and learn self-advocacy skills within the context of the educational planning process. Too often, students' perspectives have been lost because they have not had the opportunities or the skills to express their perspective within the IEP, transition, or general educational planning meetings. A first step to enabling students to express their wants and needs during these meetings is educating students about their rights and responsibilities in these areas. They can be educated about their educational rights and responsibilities under IDEA; about their civil rights under the Americans with Disabilities Act (ADA) of 1990 (PL 101-336); or, more generally, about the rights available to citizens. Instructional strategies have been developed for teaching such knowledge to students with disabilities (Wehmeyer et al., 2007).

When one is teaching students how to advocate for themselves, the focus should be on teaching students how to be assertive, how to effectively communicate their perspective (either verbally or in written or pictorial form), how to negotiate, how to compromise, and how to deal with systems and bureaucracies. Students need to be provided real-world opportunities to practice these skills. This can be done by embedding opportunities for self-advocacy within the school day, by allowing students to set up a class schedule, work out their supports with a resource room teacher or other support provider, or participate in IEP and transition meetings.

Perceptions of Efficacy and Control

People who have positive perceptions of their efficacy believe they can perform the behavior required to achieve a desired outcome (Bandura & Cervone, 2000). Furthermore,

individuals also have efficacy expectations, which are beliefs about the probability of the performance of a given behavior leading to the desired outcome. These two constructs are both necessary for the performance of the skills discussed previously. For example, if a student does not believe that he or she can perform a particular behavior, then he or she will not engage in it (i.e., if a student with a disability does not believe he or she has the requisite skills for communicating with his or her peers, then he or she will not make attempts to communicate). However, even if a student does believe he or she can perform a given behavior, but holds low expectations for the attainment of the desired result from the behavior because previous attempts to engage in the behavior have been ignored or disregarded, then he or she is still likely not to perform the behavior.

Research has shown that students with disabilities tend to have less adaptive perceptions of efficacy and outcome expectations than do students without disabilities (Wehmeyer et al., 2003). The same has been found concerning the perceptions of students with disabilities about their ability to exert control over their environment. People who believe they have the ability to exert control over their lives and outcomes tend to be described as having an internal locus of control, whereas people who perceive that others are largely controlling their lives and outcomes are described as having an external locus of control (Rotter, 1966). Students must be provided with opportunities to develop adaptive perceptions of their efficacy in performing given behaviors and their ability to exert control over their lives. By enabling students to engage in problem solving, goal setting, choice making, and decision making, they can learn that they have control over their outcomes and develop confidence in their ability to perform these behaviors and achieve their desired outcomes. Both teacher and classroom characteristics can influence students' perceptions of efficacy and control. Overly controlling environments can diminish students' perceptions of their ability to exert control and engage in actions that enable them to develop adaptive efficacy expectations. It is important for teachers to work to empower students to be active participants in their classrooms.

Self-Awareness and Self-Knowledge

For students to become more self-realizing, they must possess a reasonably accurate understanding of their strengths, abilities, unique learning and support needs, and limitations. Furthermore, they must know how to utilize this understanding to maximize success and progress. However, like perceptions of efficacy and control, self-awareness and -knowledge are not things that can simply be taught through direct instruction. Instead, students acquire this knowledge by interacting with their environment. Unfortunately, students with disabilities often learn to identify what they *cannot* do instead of what they can do. This skews students' perceptions of themselves and influences how they interact with people and systems they encounter.

Thus, it is important to promote *realistic* self-awareness and -knowledge in students with disabilities. For example, Faherty (2000) developed an approach to guide children and youth with autism spectrum disorders through the process of developing an understanding of their strengths, their abilities, and the impact of autism on their lives. The process has a number of activities that encourage students to think about their strengths and abilities and contains activities to support students to develop and reflect on how they learn, their sensory experiences, their artistic and technological abilities, their social and communication skills, their thoughts, and why they sometimes feel upset. It also helps students reflect on the people in their lives, including their school experiences.

Self-Determined Learning Model of Instruction

Like all educators, special education teachers use a variety of *teaching models*, which are defined as "a plan or pattern that can be used to shape curriculums (long-term courses of

study), to design instructional materials, and to guide instruction in the classroom and other settings" (Joyce & Weil, 1980, p. 1). Such models are derived from theories about human behavior, learning, or cognition, and effective teachers employ multiple models of teaching, taking into account the unique characteristics of the learner and types of learning. A teacher may use the role-playing model to teach social behaviors, social simulation and social inquiry models to examine social problems and solutions, assertiveness training to teach self-advocacy skills, or a training model to teach vocational skills. Likewise, special educators employ more traditional, cognitively based models of teaching, such as the concept attainment model to teach thinking skills, the memory model for increasing the retention of facts, or inductive thinking and inquiry training models to teach reasoning and academic skills. The teaching model most frequently adopted by special educators is the contingency management model, drawing from operant psychology.

The common theme across these teaching models is that they are teacher-directed. Although they provide direction for strategy and curriculum development activities that can teach components of self-determination, none adequately provides teachers a model to truly enable young people to become causal agents in their lives. The *Self-Determined Learning Model of Instruction* (SDLMI; Mithaug, Wehmeyer, Agran, Martin, & Palmer, 1998; Wehmeyer et al., 2000) was developed to address this problem and is based on the component elements of self-determination, the process of self-regulated problem solving, and research on student-directed learning. It is appropriate for use with students with and without disabilities across a wide range of content areas and enables teachers to engage students in the totality of their educational program by increasing opportunities to self-direct learning and, in the process, to enhance student self-determination.

Implementation of the model consists of a three-phase instructional process depicted in Figures 2.1, 2.2, and 2.3. Each instructional phase presents a problem to be solved by the student. The student solves each problem by posing and answering a series of four student questions per phase that students learn, modify to make their own, and apply to reach self-selected goals. Each question is linked to a set of teacher objectives. Each instructional phase includes a list of educational supports that teachers can use to enable students to self-direct learning. In each instructional phase, the student is the primary agent for choices, decisions, and actions, even when eventual actions are teacher-directed.

The student questions are constructed to direct the student through a problem-solving sequence in each instructional phase. Teachers implementing the model teach students to solve a sequence of problems to construct a means-ends chain—a causal sequence—that moves them from where they are (an actual state of not having their needs and interests satisfied) to where they want to be (a goal state of having those needs and interests satisfied). To answer the questions in this sequence, students must regulate their own problem solving by setting goals to meet needs, constructing plans to meet goals, and adjusting actions to complete plans. The questions differ from phase to phase but represent identical steps in the problem-solving sequence. That is, students answering the questions must

1. Identify the problem
2. Identify potential solutions to the problem
3. Identify barriers to solving the problem
4. Identify consequences of each solution

These steps are the fundamental steps in any problem-solving process.

Because the model itself is designed for teachers to implement, the language of the student questions is not written to be understood by every student, nor does the model assume that students have life experiences that enable them to fully answer each question. Some students will learn and use all 12 questions as they are written. Other students will need to have the questions rephrased to be more understandable. Still other students,

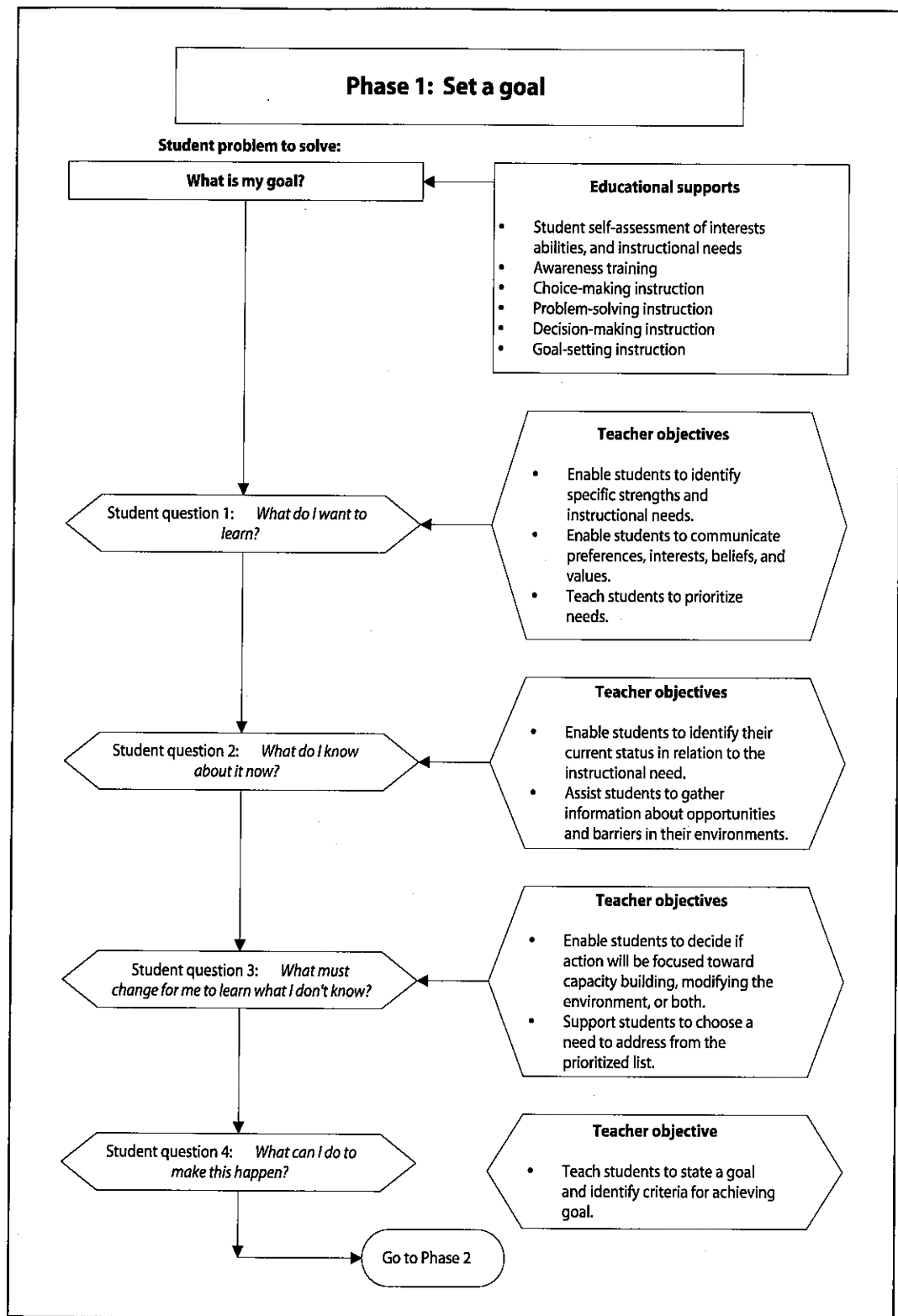


Figure 2.1. Phase 1 of the Self-Determined Learning Model of Instruction.

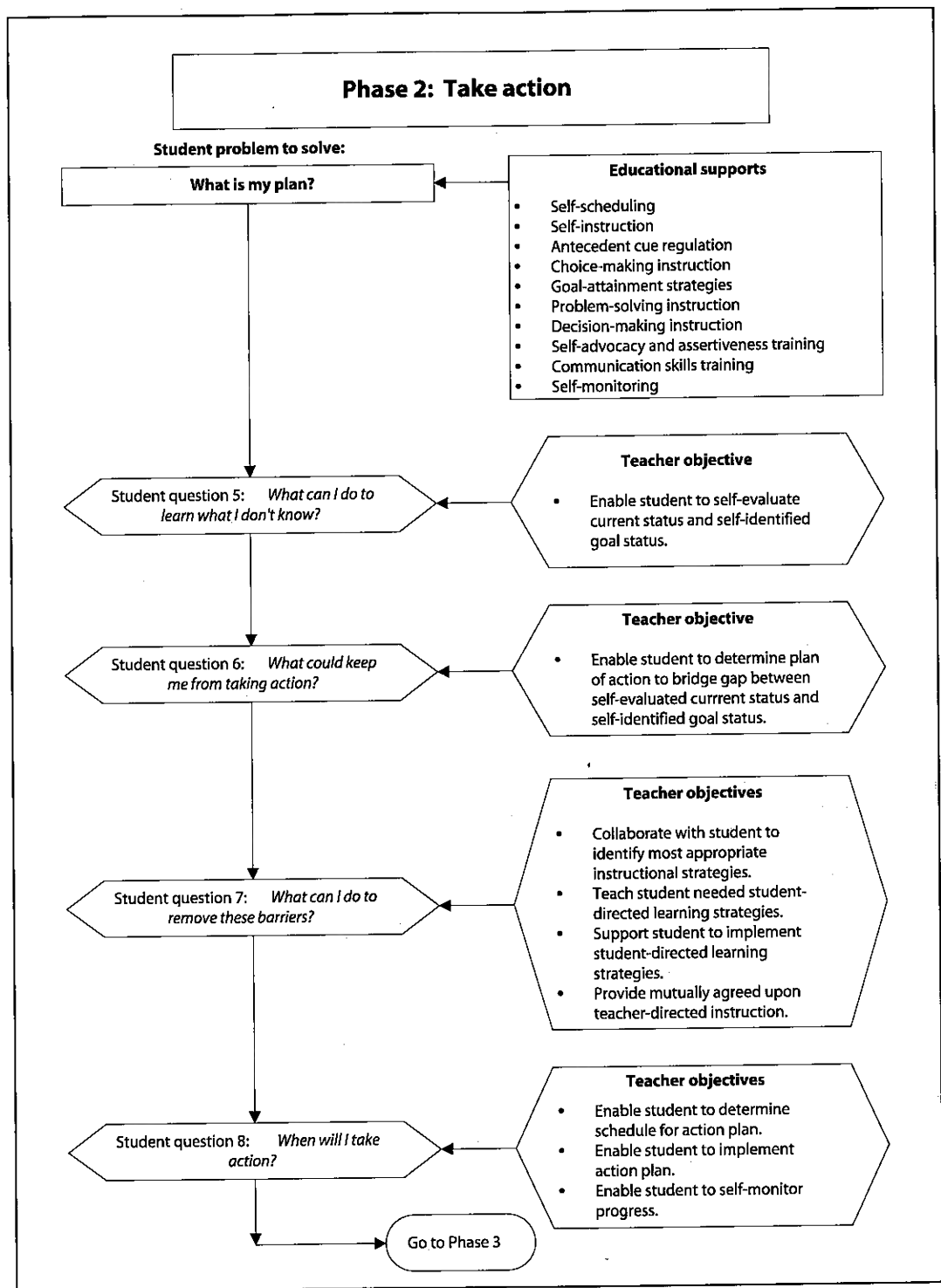


Figure 2.2. Phase 2 of the Self-Determined Learning Model of Instruction.

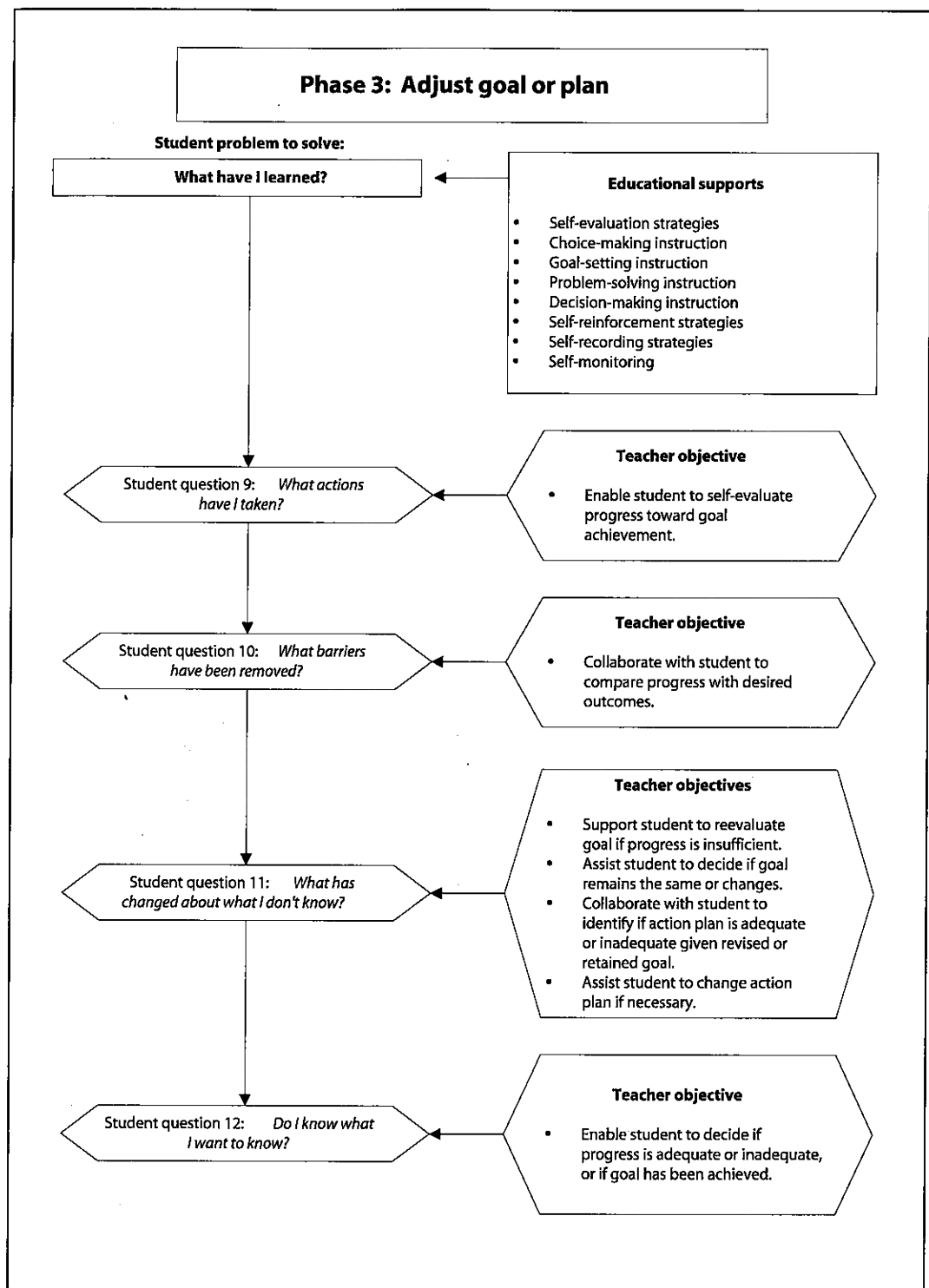


Figure 2.3. Phase 3 of the Self-Determined Learning Model of Instruction.